

To: Devon Achebe, VP of Customer Retention

From: Jared Mendez

Subject: Recommendation for a Retention Pilot

I recommend running a small randomized retention pilot using boosted trees to select the highest-risk 20% of customers rather than launching the campaign broadly. On the final test, boosted trees achieved an AUC of 0.849665, compared with 0.847207 for logistic regression and 0.737265 for the contract rule. Its 281-customer contact list had a 69.395% historical churn rate, versus 26.544% overall and 39.858% for the contract-rule list.

Under the fictional planning assumptions, projected value per 1,000 contacts is $-\$1,619.93$ at a 10% save rate, $\$670.11$ at 15%, and $\$2,960.14$ at 20%. The break-even save rate is 13.5369%.

These figures are hypothetical, not measured profit. Predicted churn does not establish that an offer causes customers to stay, and trees' AUC advantage over logistic regression is uncertain. Randomly assign selected customers to offer and control groups, then measure incremental retention and full campaign costs before considering a larger rollout.